

Yi Ding

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RESEARCH INTERESTS

- My research interests lie at the intersection of **mobile computing**, **wireless sensing**, and **machine learning**. I focus on building **perceptive mobile AI** systems that can sense and understand the *physical environment* and *human behavior*, enabling intelligent and context-aware applications.

EMPLOYMENT

Assistant Professor in Computer Science <i>University of Texas at Dallas, Richardson, TX, USA</i>	Aug. 2023 - Now
Postdoctoral Associate <i>Massachusetts Institute of Technology (MIT), MA, USA</i> Advisor: Prof. Alex “Sandy” Pentland (MIT Media Lab, MIT IDSS)	Oct. 2022 - Jul. 2023

EDUCATION

Ph.D. of Computer Science <i>University of Minnesota, USA</i> Advisor: Prof. Tian He (IEEE/ACM Fellow)	Sep. 2016 - Oct. 2022
Visiting Student <i>Rutgers University, USA</i> Host: Prof. Desheng Zhang	Feb. 2022 - Oct. 2022
Master of Control Science and Engineering <i>Zhejiang University, China</i> Advisor: Prof. Zuhua Xu	Sep. 2013 - Jun. 2016
Visiting Student <i>IMT Lucca, Italy</i> Host: Prof. Alberto Bemporad (IEEE Fellow)	Aug. 2015 - Oct. 2015
Bachelor of Control Science and Engineering <i>Zhejiang University, China</i>	Sep. 2009 - Jun. 2013 with Outstanding Thesis Award

HONORS & AWARDS

- CPS Rising Star, CPS Rising Stars Workshop, University of Virginia, 2022
- Outstanding Paper Award, IEEE RTSS, 2021
- Outstanding Thesis Award, Zhejiang University, 2013

PUBLICATIONS

Highlights: 36 papers, featuring 23 papers in top-tier conferences, including NSDI, SIGCOMM, MobiCom, UbiComp, RTSS, VLDB, SIGKDD, and WWW, with 1 outstanding paper award. The authors with * are the students working primarily with me.

Conference

- [C29] Anlan Yu*, Zhiqing Hong, Jinkun Li, Shuting Pan, Haotian Wang, **Yi Ding**, Daqing Zhang
MobiCom’26 *Experience: Unveiling GNSS-RTK in the Wild: A 2.5-Year Nationwide Study via Last-Mile Delivery*
 Proceedings of the 32th Annual International Conference on Mobile Computing and Networking
- [C28] Fangwei Zhang, Yimeng Liu, Franklin Wu, **Yi Ding**, Zhichao Cao, Ruofeng Liu
MobiCom’26 *UltraLEGO: A Modular Ultra-Wideband Phased Array Platform for Wireless Sensing and Localization*
 Proceedings of the 32th Annual International Conference on Mobile Computing and Networking
- [C27] Ziqi Liu*, Pei Zeng*, **Yi Ding**
WWW’26 *Predictability-Aware Compression and Decompression Framework for Multichannel Time Series Data with Latent Seasonality*
 ACM The Web Conference 2026
- [C26] Hua Yan, Heng Tan, **Yi Ding**, Pengfei Zhou, Vinod Namboodiri, Yu Yang
UbiComp’26 *Large Language Model-Guided Semantic Alignment for Human Activity Recognition*
IMWUT In ACM International Joint Conference on Pervasive & Ubiquitous Computing
- [C25] Shanshan Wang*, S. Duan, S. Zhong, Z. Hong, Z. Zhou, H. Lin, W. Zuo, Desheng Zhang, **Yi Ding**
CIKM’25 *Fraudulent Delivery Detection with Multimodal Courier Behavior Data in Last-Mile Delivery*
 34rd ACM International Conference on Information and Knowledge Management
- [C24] Zhiqing Hong, Weibing Wang, Anlan Yu, S. Zhong, H. Wang, **Yi Ding**, Tian He, Desheng Zhang
MobiCom’25 *Experience Paper: Nationwide Human Behavior Sensing in Last-mile Delivery*
 Proceedings of the 31th Annual International Conference on Mobile Computing and Networking
- [C23] Feng Lyu, Lijuan He, Mingliu Liu, Sijing Duan, H. Wu, J. Zhou, **Yi Ding**, Zaixun Ling, Yibo Cui
MobiCom’25 *Auto-UIT: Automated UAV Inspection Trajectory Generation from Noisy Sparse 3D Point Cloud*
 Proceedings of the 31th Annual International Conference on Mobile Computing and Networking
- [C22] Zhiqing Hong, Yiwei Song, Zelong Li, Anlan Yu, S. Zhong, **Yi Ding**, Tian He, Desheng Zhang
SIGKDD’25 *Generalizable On-device HAR with Pretrained LLMs for Last-mile Delivery*
 Proceedings of the 31th ACM SIGKDD Conference on Knowledge Discovery and Data Mining
- [C21] Xiao Yan*, **Yi Ding**
FmSys’25 *Are We There Yet? A Measurement Study of Efficiency for LLM Applications on Mobile Devices*
 International Workshop on Foundation Models for Cyber-Physical Systems & Internet of Things
- [C20] Wenjun Lyu, S. Zhong, G. Yang, H. Wang, **Yi Ding**, S. Wang, Y. Liu, T. and Desheng Zhang
WWW’25 *InCo: Exploring Inter-Trip Cooperation for Efficient Last-mile Delivery*
 ACM The Web Conference 2025
- [C19] Anlan Yu*, X. Wang, P. Wang, J. Li, X. Ma, Z. Hong, Haotian Wang, **Yi Ding**, Daqing Zhang
PerCom’25 *FineSat: Enhancing GNSS Signals for High-precision Sensing*
 2025 IEEE International Conference on Pervasive Computing and Communications (PerCom)

- [C18]
CIKM'24 Guang Yang*, Meiqi Tu, Z. Li, J. Hang, T. Liu, R. Liu, **Yi Ding**, Yu Yang and Desheng Zhang
Behavior-Aware Hypergraph Convolutional Network for Illegal Parking Prediction with Multi-Source Contextual Information
33rd ACM International Conference on Information and Knowledge Management
- [C17]
UbiComp'24
IMWUT Zhiqing Hong, Zelong Li, S. Zhong, W. Lyu, Haotian Wang, **Yi Ding**, Tian He, Desheng Zhang.
CrossHAR: Generalizing Cross-dataset HAR via Hierarchical Self-Supervised Pretraining
In ACM International Joint Conference on Pervasive & Ubiquitous Computing
- [C16]
UbiComp'24
IMWUT Zhiqing Hong, Haotian Wang, **Yi Ding**, Guang Wang, Tian He, Desheng Zhang.
SmallMap: Low-cost Community Road Map Sensing with Uncertain Delivery Behavior
In ACM International Joint Conference on Pervasive & Ubiquitous Computing
- [C15]
VLDB'24 Sijing Duan, Feng Lyu, Xin Zhu, **Yi Ding**, H. Wang, Desheng Zhang, Xue Liu, Y. Zhang, Ju Ren
VeLP: Vehicle Loading Plan Learning from Human Behavior in Nationwide Logistics System
International Conference on Very Large Data Bases
- [C14]
SIGSPATIAL'22 Zhiqing Hong, Guang Wang, W. Lyu, B. Guo, **Yi Ding**, H. Wang, S. Wang, Y. Liu, D. Zhang
CoMiner: Nationwide Behavior-driven Unsupervised Spatial Coordinate Mining
ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems
- [C13]
SIGKDD'22
195/753=26% Wei Liu*, **Yi Ding**, Shuai Wang, Yu Yang, Desheng Zhang.
Para-Pred: Addressing Heterogeneity for City-Wide Indoor Status Estimation
Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining
- [C12]
UbiComp'22 Baoshen Guo, Weijian Zuo, Shuai Wang, W. Lyu, Z. Hong., **Yi Ding**, Tian He, Desheng Zhang.
WePos: Weak-supervised Indoor Positioning with Unlabeled WiFi for On-demand Delivery
In ACM International Joint Conference on Pervasive & Ubiquitous Computing
- [C11]
MobiCom'22
16/91=18% Pengfei Zhou, **Yi Ding**, Yang Li, Mo Li, Guobin Shen, Tian He.
Experience: Adopting Indoor Outdoor Detection in On-demand Food Delivery Business
In Annual International Conference on Mobile Computing and Networking
- [C10]
UbiComp'22 **Yi Ding**, Dongzhe Jiang, Yu Yang, Yunhuai Liu, Desheng Zhang, Tian He.
P2-Loc: A Person-2-Person Indoor Localization System in On-Demand Delivery
In ACM International Joint Conference on Pervasive & Ubiquitous Computing
- [C09]
UbiComp'22 **Yi Ding**, Dongzhe Jiang, Yunhuai Liu, Desheng Zhang, Tian He.
SmartLOC: Indoor Localization with Smartphone Anchors for On-Demand Delivery
In ACM International Joint Conference on Pervasive & Ubiquitous Computing
- [C08]
ICPADS'21 Zhigang Dai, Wenjun Lyu, **Yi Ding**, Yiwei Song
OPTI: Order Preparation Time Inference for On-demand Delivery
In IEEE International Conference on Parallel and Distributed System
- [C07]
RTSS'21 Baoshen Guo, Shuai Wang, **Yi Ding**, Guang Wang, Suining He, Desheng Zhang, Tian He.
Concurrent Order Dispatch for Instant Delivery with Time-Constrained Actor-Critic Reinforcement Learning
In IEEE Real-Time Systems Symposium (**Outstanding Paper Award**)
- [C06]
UbiComp'21 Dongzhe Jiang, **Yi Ding**, Hao Zhang, Yunhuai Liu, Tian He, Yu Yang, Desheng Zhang.
ALWAES: an Automatic Outdoor Location-Aware Correction System
In ACM International Joint Conference on Pervasive & Ubiquitous Computing
- [C05]
UbiComp'21 **Yi Ding**, Baoshen Guo, Lin Zheng, M. Lu, Desheng Zhang, S. Wang, Sang H. Son, Tian He.
A City-Wide Crowdsourcing Delivery System with Reinforcement Learning
In ACM International Joint Conference on Pervasive & Ubiquitous Computing

- [C04] **Yi Ding**, Yu Yang, Wenchao Jiang, Yunhuai Liu, Tian He, Desheng Zhang.
SIGCOMM'21 55/241=22%
Nationwide Deployment and Operation of a Virtual Arrival Detection System in the Wild
 In the Annual Conference of the ACM Special Interest Group on Data Communication
- [C03] **Yi Ding**, Ling Liu, Yu Yang, Yunhuai Liu, Desheng Zhang, Tian He.
NSDI'21 19/114=17%
A Lifetime Story of a 3-Year-Old Operational Wireless Beacon System in the Wild
 In USENIX Symposium on Networked Systems Design and Implementation
- [C02] Yu Yang, **Yi Ding**, D. Yuan, G. Wang, X. Xie, Yunhuai Liu, Tian He, Desheng Zhang.
MobiCom'20 62/384=16%
Transparent Indoor Localization with Uncertain Human Participation for Instant Delivery
 In Annual International Conference on Mobile Computing and Networking
- [C01] Yan Zhang, Yunhuai Liu, **Yi Ding**, G. Li, N. Chen, H. Zhang, Tian He, Desheng Zhang.
UbiComp'19
Route Prediction for Instant Delivery
 In ACM International Joint Conference on Pervasive & Ubiquitous Computing

Journal

- [J07] Anlan Yu*, Xujun Ma, Z. Hong, P. Wang, X. Wang, Jinkun Li, Haotian Wang, **Yi Ding**, Daqing Zhang
TMC'26
Beyond Navigation: Unlocking the High-precision Sensing Potential of GNSS Signals
 In IEEE Transactions on Mobile Computing
- [J06] Sijing Duan, Feng Lyu, Shanshan Wang, **Yi Ding**, Xiaohao He, Yaoxue Zhang
TMC'25
Exploring Cellular User Re-Identification Risks with Networking Behaviors Analysis and Modeling
 In IEEE Transactions on Mobile Computing
- [J05] Zhiqing Hong, G. Wang, W. Lyu, B. Guo, **Yi Ding**, H. Wang, S. Wang, Yunhuai Liu, Desheng Zhang
TKDE'24
Nationwide Behavior-Aware Coordinates Mining from Uncertain Delivery Events
 In IEEE Transactions on Knowledge and Data Engineering
- [J04] Shuai Wang, Baoshen Guo, **Yi Ding**, Guang Wang, Suining He, Desheng Zhang, Tian He
TMC'23
Time-Constrained Actor-Critic Reinforcement Learning for Concurrent Order Dispatch in On-demand Delivery
 In IEEE Transactions on Mobile Computing
- [J03] Zhigang Dai, Wenjun Lyu, **Yi Ding**, Yiwei Song, Yunhuai Liu
TOSN'23
OPTI: Order Preparation Time Inference for On-demand Delivery
 In ACM Transactions on Sensor Network
- [J02] **Yi Ding**, Yu Yang, Wenchao Jiang, Yunhuai Liu, Tian He, Desheng Zhang.
ToN'22
Nationwide Deployment and Operation of a Virtual Arrival Detection System in the Wild
 In IEEE/ACM Transactions on Networking
- [J01] **Yi Ding**, Ling Liu, Yu Yang, Yunhuai Liu, Desheng Zhang, Tian He.
ToN'21
A Lifetime Story of a 3-Year-Old Operational Wireless Beacon System in the Wild
 In IEEE/ACM Transactions on Networking

GRANTS

- UTD Seed: SPIRe: An Adaptive Edge-Intelligent Digital Twin for Resilient Lunar Nuclear Energy Systems, 06/2026 - 05/2027, Budget: \$60,000, Role: Co-PI, Share: \$25,000
- NSF CISE NeTS CSR, Leveraging Satellite Signals for Ubiquitous Mobile Wireless Sensing, 08/2025 - 07/2028, Budget: \$400,000, Role: PI, Share: \$230,957

PROFESSIONAL ACTIVITIES

Invited Talks

- Invited Talk at CPS-IoT Week 2025 Sensor S&P workshop, U.S., May 2025
- Invited Talk at the University of North Dakota, U.S., August 2024
- Invited Talk at MIT (Media Lab), U.S., August 2022
- Invited Talk at Kargo (Logistics Startup), U.S., July 2022
- Invited Talk at Location-Based Service Team, ByteDance, China, June 2022
- Invited Talk at Zhongshan Workshop, China Computer Federation (CCF), Nanjing, China, December 2021

Organizers

- ACM SIGSPATIAL SpatialConnect Workshop 2026, 2025
- IEEE Cloud Summit (Track Chair) 2026, 2025

Associate Editors

- Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)

Program Committee for Conferences

- ACM MobiSys 2025
- ACM Special Interest Group on Data Communication Posters and Demos (SIGCOMM'26,'25,'24,'23)
- IEEE PerCom WIP 2026
- ACM SenSys IoT-MMLM Workshop 2024
- ACM MobiCom Picasso Workshop 2024
- International Conference on Algorithms and Architectures for Parallel Processing (ICA3PP'24, '23)

(External) Reviewer

- IEEE Transactions on Mobile Computing (TMC)
- ACM Transactions on Sensor Networks (TOSN)
- ACM Transactions Internet Things (TIOT)
- IEEE Transactions on Network Science and Engineering (TNSE)
- IEEE Transactions on Cloud Computing (TCC)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Intelligent Transportation Systems
- ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (SIGKDD)
- International Conference on Learning Representations (ICLR)
- Annual AAAI Conference on Artificial Intelligence
- ACM Conference on Information and Knowledge Management (CIKM)

Panelist

- NSF CPS, 2024
- NSF SBIR, 2024
- NSF SCC, 2024, 2025

University and Department Service

- UT Dallas Advisory Committee on Research 2025-2026
- CS Department Admission Committee 2023-2026
- CS Department K-12 Outreach Committee 2023-2026

TEACHING

- Fall'26, Spring'26, Fall'25, Spring'25: Research, Inquiry, Design Experience (RIDE) (UTD ECS1192/2192/3192/4192)
- Fall'26, Fall'25: Special Topics in CS: Machine Learning in Mobile Computing (UTD CS 6301)
- Spring'25, Fall'24, Spring'24: Computer Networks (UTD CS/CE 4390)
- Fall'23: Special Topics in CS: Data Science for Smart Cities (UTD CS 6301)

MENTORING

- **PhD:** Ziqi Liu (2024-), Xiao Yan (2024-), Tianyu Fang (2026-)
- **Master Students:** Tushar Wani (UTD Independent Study, Spring 26), Buvana Seshathri (UTD Independent Study, Spring 26), Lokesh Balasubramaniam (UTD Master Thesis, Spring 26),
- **Undergraduate Students:** Sanskriti Aripineni (UTD RIDE, Spring'25-Fall'26), Rishi Simhadri (UTD RIDE, Fall'25-Fall'26), Shivesh Gupta (UTD URAP, Summer 26)
- **High School Students:** William Zhou (Summer 26, Plano West Senior High, Plano, Texas) Sai Deepthi Puligadda (Summer 26, Rouse High School, Leander, Texas) Mayukha Siripuram (Summer 26, Centennial High School, Frisco, Texas)

GROUP ALUMNI

- **Visiting PhD:** Tong Liu (2024-2025)
- **Master Students:** Pei Zeng (2024-2025, Now in LexisNexis), Chitsein Alexander Dixon Htun (UTD URAP, Summer 25, Now PhD at UIUC)
- **Undergraduate Students:** Khoa A Tran, Lucky Srini (UTD RIDE, Fall'25, Spring'26), Dishita Thathapuram, Hishita Rakesh Shah (UTD RIDE, Spring'25), Anindo Majumdar (2024, Now M.S. at UTD)
- **High School Students:** Aditi Chhawacharia (Summer 25, Walnut Grove High School, Prosper, Texas), Sanjana Mothukuri (Summer 25, Panther Creek High School, Frisco, Texas), Vidushi Nalakonda (Summer 25, Coppell High School, Coppell, Texas), Roger Wang (Summer 25, Centennial High School, Frisco, Texas, Now at UT Austin)

DATA-SET RELEASE

- aBeacon: collected from a citywide physical beacon system, including the data of Bluetooth, manual report, and location from 31,131 couriers at 2,466 merchant locations in one month. [\[Link\]](#)
- VALID: collected from a nationwide virtual beacon system, including the data of Bluetooth, manual report, and manual feedback from 55,000 couriers at 113,000 merchant locations in ten cities in one month. [\[Link\]](#)
- RL-Dispatch: collected from a citywide crowdsourcing delivery system, including the on-demand delivery order data in one month. [\[Link\]](#)
- P2Loc: collected in a encounter-based localization project, including the BLE-based indoor encounter data and reported data of 176 couriers at 54 merchant locations in a mall in one month.. [\[Link\]](#)

MEDIA EXPOSURE

- MIT Technology Review
- Alibaba Technology

PATENTS

- Embedded MPC method based on convergence depth control [\[Link\]](#)
- Indoor positioning method and apparatus, electronic device and computer readable storage medium. [\[Link\]](#)

- Information processing method and device. [[Link](#)]
- Position state acquisition method and device, readable storage medium and electronic equipment. [[Link](#)]
- Order processing method and device, storage medium and electronic equipment [[Link](#)]
- Beacon association method and device. [[Link](#)]
- Task execution method and device based on gesture interaction. [[Link](#)]